Emergency Training Simulator POC

Submission Packet

Full project vision

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Source: docs/full-project-vision.md

Prepared for review, discussion, and project handoff.

Markdown stays editable. This PDF is the clean reading copy.

Boundary: this is still a POC, not operational or clinical guidance.

Full project vision

This document summarizes the full project goal from docs/reference/full-project-overview.docx.

The full project is bigger than the current POC.

The current repo now combines the deterministic foundation with hosted room planning, Blender/Unreal chair workflows and implemented asset automation. See current status for evidence and outstanding live checks.

For a side-by-side map of full-system goals, current repo status, open gaps, and safe next experiments, see full-system-gap-map.md.

North-star goal

The long-term goal is an AI-driven, real-time adaptive emergency-response training platform for firefighters and EMTs.

The full system would combine:

- agentic AI scenario generation
- real-time scenario adaptation
- immersive training environments
- physical mannequins
- multimodal trainee sensing
- structured verification and validation
- cloud-enabled hub-and-spoke deployment
- post-scenario review and continuous improvement

Full-system loop

At full scale, the intended loop is:

```
trainee speaks, moves, and performs actions
-> sensing captures behavior
-> AI interprets trainee actions
-> scenario state adapts in real time
-> validation checks safety and correctness
-> immersive environment updates
-> session data is logged
-> review and learning loop improve future training
```

Deployment vision

The full project describes a cloud-enabled hub-and-spoke model.

In that future architecture:

- Virginia Tech can act as a regional hub
- pilot spoke sites may include Alexandria Fire Department and the City of Philadelphia
- cloud AI services generate and adapt scenarios
- regional hubs support coordination, caching, resilience, and lower-latency operation
- distributed sites can share consistent training intelligence without each site needing heavy local infrastructure

This is future architecture. The current repo does not implement hub-and-spoke networking.

AI vision

The full system eventually includes AI and agentic AI that can:

- generate scenario content
- adapt emergency events in real time
- interpret trainee behavior
- update patient state
- update environmental hazards
- support post-scenario review

The room studio calls ARC's hosted model. Description-based SDXL/Hunyuan asset generation is implemented but still needs its first live automated couch run. These authoring/generation paths do not implement validated AI adaptation during a trainee session.

For now, deterministic code stands in for future adaptation so the backend loop can be tested and trusted first.

Validation vision

The full project includes a verification and validation framework for AI outputs.

In the full system, validation may compare generated patient states, scenario changes, and intervention outcomes against structured sources such as:

- medical dictionaries
- treatment protocols
- dosage guidelines
- expected outcomes
- validated department-provided materials

The current repo does not claim clinical validity or protocol authority.

Current validation means only:

Did this action follow the rules of this toy deterministic POC scenario?

Immersive environment vision

The full system may eventually include immersive rooms with:

- multi-projector setups
- projection mapping
- mannequin localization
- projected patient/environment changes
- synchronized audio/visual updates
- physical mannequin integration

The repo has generated-chair evidence, Blender room assembly and Unreal 5.8 scene scripts/Chaos demonstrations. Projection mapping, complete mixed-furniture environments and integrated trainee gameplay remain future work.

Sensing vision

The full system may eventually observe trainee behavior through:

- voice input
- computer vision
- pose and gesture recognition

- motion tracking
- wearable or environmental sensors
- procedural action detection

The current repo does not implement multimodal sensing.

Manual or scripted trainee actions are used as safe stand-ins.

Accessibility and scaling vision

The full project aims to support both high-fidelity immersive deployments and lower-cost configurations.

The larger goal is to make advanced training more accessible to departments with different budgets, staffing levels, and locations.

Relationship to this repo

This repo is not the final platform.

It currently proves the smallest trustworthy software loop:

```
manual/stubbed action
-> backend receives event
-> deterministic scenario engine updates state
-> POC validation checks rule conformance
-> client receives state
-> JSONL log is written
-> replay/review explains what happened
```

That loop matters because the full system will eventually need the same foundation, even when AI, Unreal, sensing, projection, networking, and richer validation are added later.

Current build strategy

Because the project is currently being advanced by one solo developer, the full vision is being treated as a north star rather than a single immediate build.

The practical strategy is:

1. prove the backend loop
2. make the session explainable through replay and review
3. make Unreal display backend-owned state
4. add simple visuals only after the display path works
5. review assets manually before trusting generated content
6. validate the new asset/engine lane and mixed rooms, then extend sensing, adaptation, domain validation and deployment through measured checkpoints

This keeps the project realistic without shrinking the long-term goal.

Important boundary

The full project vision includes future capabilities that are not implemented here yet.

Do not describe the current repo as having:

- validated AI adaptation during active training
- clinical validation
- official EMT/firefighter protocol authority
- hub-and-spoke deployment

- projection mapping
- multimodal sensing
- full Unreal gameplay
- arbitrary automated semantic mesh repair
- calibrated real-world physics or validated training effectiveness
- cloud deployment

The current POC is valuable because it is smaller, testable, replayable, and honest.